

LeetCode 237: Delete Node in a Linked List

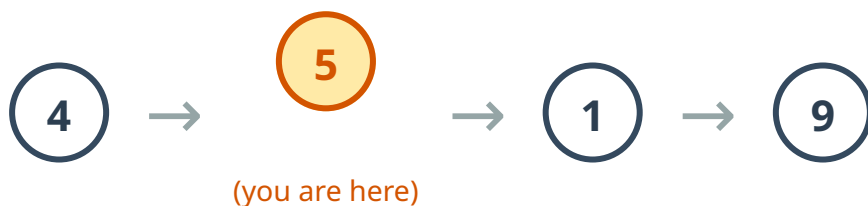
Frequently Asked: Cisco, NVIDIA, Meta

The Twist: We are *not* given the head pointer of the linked list. Instead, we are currently standing on the node to be deleted itself.

The Problem: Because we don't know head, we cannot traverse the list from the beginning to find the prev (previous) node, which is fundamentally required for standard node deletion.

Visual Walkthrough

Example: We are asked to remove node '5'



Step 1: Copy the value of the next node into the current node.



Step 2: Now simply delete the next node (since you have the current node acting as its `prev` to re-establish the connection).



Code Implementation

```
class Solution {
public:
    void deleteNode(ListNode* node) {

        node->val=node->next->val;
        ListNode* tbd=node->next;
        node->next=node->next->next;
        delete(tbd);

    }
};
```

Complexity Analysis

Time Complexity: $O(1)$

Space Complexity: $O(1)$

⚠ **Important Note:** This approach works perfectly because LeetCode guarantees that the given node to be deleted is *not* a tail node.